

What Syndrome Data Can and Cannot Detect: Accessible-Observable Sequential Change Detection for Topological Quantum Error Correction

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Abstract—Surface-code monitors often compress a detector stream to a defect rate, although drift can preserve every detector marginal and the complete count distribution while changing spatial or temporal correlations. We formulate the resulting limitation at the process level: if two hypotheses induce the same law on the entire accessible observation path, every stopping rule adapted to that path has the same distribution under both hypotheses. One-round equality alone is insufficient when dependence changes. We then construct a periodic two-endpoint detector model in which (i) spatial chain-length drift is exactly count-blind but is identified by two signed translation correlations, (ii) temporal persistence preserves the complete one-cycle syndrome law but changes adjacent-pair likelihoods, and (iii) a closed logical cycle is pathwise syndrome-invisible. The spatial likelihood reduces exactly to three translation statistics, and the temporal pair likelihood has a closed form. A predictable, bounded observable score is embedded in the established Shiryayev–Roberts/e-detector construction; under an exact null it has an average-run-length guarantee. Locked identity checks give zero count total variation and numerical likelihood errors below 9×10^{-15} . Separate audits find a 4.20–4.23 copy-RMSE penalty for local-Pauli shadows versus native all- Z readout on declared ZZ observables, and a controlled Stim/PyMatching experiment reduces the distance-seven correlated-regime logical-error rate from 0.01526 to 0.01120 when the injected post-change correlation model is known. That circuit result tests decoder utility, not change detection. In the cycle-fair locked sequential test, VAOC has restart delays 540.30 versus 470.13 cycles for the named validation-trained logistic comparator at the spatial endpoint; at the temporal endpoint they are 959.36 versus 968.54. The corresponding VAOC-minus-logistic differences are 70.18 and -9.18 cycles, but neither predeclared hypothesis passes the simultaneous distribution-free/Holm criterion. Thus the experiment supports no VAOC advantage over the named comparator on either task or overall.

Index Terms—accessible observables, change detection, e-detectors, quantum error correction, surface code, syndrome correlations

I. INTRODUCTION

SURFACE-CODE syndrome data are valuable not only for decoding but also for estimating and tracking noise. Existing work learns detector-error-model (DEM) events from repeated syndromes, performs Bayesian noise inference, and adapts decoding graphs under drift and correlated faults [1]–[4]. The first question is therefore not whether syndromes can reveal drift in general. It is which drift remains identifiable

Run 5 working paper. Exact statements, controlled-model results, circuit-level results, and absent hardware evidence are labeled separately. All numerical claims are backed by frozen configurations, raw records, and manifests in the accompanying repository.

after a declared compression, locality restriction, measurement policy, and history budget.

Run 4 of the accompanying repository gave an exact static example: a sub-distance observer is blind to a toric-code flux sector, whereas a noncontractible Wilson loop resolves it. That local blindness and the logical role of closed chains are established topological-code/QEC structure [5]–[8]; ribbon-operator discovery is also prior art [9]. Run 5 moves from static sector classification to a practical sequential question: can correlation drift be detected at fixed syndrome and false-alarm budgets when rate/count statistics are provably blind?

The paper makes four bounded contributions.

- 1) It states an accessible-*process* no-go certificate. Equality of the full accessible path laws, not merely one-round marginals, is the condition for sequential impossibility.
- 2) It gives exact controlled spatial, temporal, and logical separations. For the spatial family, the complete count process is invariant while $(\bar{z}, g_1, g_2)$ is sufficient; for the temporal family, one-cycle laws are equal while adjacent-pair laws differ.
- 3) It places predictable raw and variance-aware observable contrasts in an established bounded e-factor/Shiryayev–Roberts construction, clearly separating one-shot mean-gap, variance-sensitive, and likelihood-growth objectives.
- 4) It reports locked identifiability, offline diagnostic, measurement-policy, sequential, scaling, and circuit-level decoder-utility audits. The predeclared named-comparator test is negative.

This is neither a claim of the first QEC drift detector nor a new covariance test. Page’s CUSUM, Hotelling whitening, Kelly log growth, and e-detectors are foundations [10]–[13]. Classical shadows and shadow-based sequential quantum change detection are also established [14], [15]. The contribution is a reproducible information-access hierarchy with negative certificates, matched witnesses, and explicit evidence boundaries.

II. PRIOR ART AND EVIDENCE BOUNDARY

A. Restricted information and sequential detection

Helstrom theory is the unrestricted binary quantum-measurement ceiling for fixed states, priors, and loss [16]. Restricted measurement families induce different distinguishability norms [17], and group-symmetric quantum hypothesis testing is established [18]. Most Run 5 experiments begin

TABLE I
EVIDENCE LEVELS AND PERMITTED CONCLUSIONS.

Level	Meaning in this paper
Exact statement	Proof under the displayed probability/QEC model.
Controlled model	Exact likelihood or Monte Carlo from the declared phenomenological family.
Circuit level	Stim/DEM samples and PyMatching for the specified simulated circuit and injected channel.
Hardware	Physical-device samples; none are used in Run 5.

after stabilizer measurement and receive a classical syndrome stream. Helstrom is therefore a conceptual upstream ceiling, not an implemented baseline for every experiment.

The variance-aware direction $V^{-1}\delta$ is classical Fisher/Hotelling geometry, not an AOC invention [11]. Its unrestricted quantum tangent-space analogue is tied to symmetric logarithmic derivatives and quantum Fisher information [19]. Likewise, the average-run-length result below specializes the e-detector framework of Shin, Ramdas, and Rinaldo [13]; it is not a new general sequential-testing theorem.

B. Four evidence levels

Table I fixes the vocabulary used throughout. “Exact” means exact under stated mathematical assumptions, not physically complete. The controlled finite-mixture model omits coherent faults, leakage, device scheduling, and many calibration mechanisms. Stim samples are circuit-level stabilizer simulations [20], and PyMatching is an established MWPM decoder [21], [22]. Neither package turns simulated data into hardware evidence.

III. ACCESSIBLE-PROCESS THEORY

A. The correct sequential no-go statement

Let $Y = (Y_1, Y_2, \dots)$ be the full observation stream. An access policy $\mathcal{H}$ maps it to $Z = \mathcal{H}(Y)$; this map may retain per-round counts, a finite history, local correlation features, or a separately costed audit. Denote the path laws by $P_j^Z = \mathcal{H}_\# P_j$, $j \in \{0, 1\}$.

Theorem 1 (accessible-process no-go). If $P_0^Z = P_1^Z$, then every randomized stopping rule adapted only to the accessible filtration has the same stopping-time law under P_0 and P_1 . In particular, for every horizon T ,

$$P_1(\tau \leq T) = P_0(\tau \leq T). \quad (1)$$

Proof: Let U be the detector’s data-independent random seed. Any such stopping rule is a measurable function $\tau = \phi(Z, U)$. The assumption implies $P_0^Z \otimes P^U = P_1^Z \otimes P^U$. Applying the measurable map ϕ to both sides gives equality of the stopping-time pushforwards and hence (1). ■

Corollary 1 (iid compression). Suppose $Z_t = h(Y_t)$ and the accessible rounds are iid under each hypothesis. If $h_\# P_0^{(1)} = h_\# P_1^{(1)}$, their path laws are equal and Theorem 1 applies.

Proof: The two accessible path measures are products of the same one-round pushforward. ■

The iid condition is essential. The temporal experiment below deliberately keeps the complete one-cycle law fixed while changing persistence. A history-aware detector then has power because the accessible *path* laws are not equal. Any statement that infers sequential blindness from one-round marginal equality alone would be false.

Data processing provides the quantitative companion

$$\text{TV}(\mathcal{H}_\# P_0, \mathcal{H}_\# P_1) \leq \text{TV}(P_0, P_1). \quad (2)$$

Thus enriching the accessible algebra or history can reveal information that compression erased, but post-processing a fixed accessible stream cannot recreate it.

B. Logical-cycle pathwise certificate

For a chain complex with syndrome boundary map ∂ , add a closed chain $c \in \ker \partial$ to an error chain E . Then

$$\partial(E + c) = \partial E. \quad (3)$$

A coupling that applies the same modification to each selected path therefore gives identical complete syndrome histories, while a homologically nontrivial c can flip a logical/Wilson audit. This is an executable instance of known QEC/local-topological-order structure, not a new toric-code or Wilson theorem. Gauge-region algebras can also have nontrivial centers and need not factor naively [23], so future gauge applications must state the accessible algebra rather than assume a tensor-product region.

IV. EXACT CONTROLLED DETECTOR MODEL

A. Spatial correlation drift

Consider an $L \times L$ periodic check lattice with $m = L^2$ detectors. In one round, $J \sim \text{Bernoulli}(\eta)$. If $J = 0$, the clean syndrome is zero. If $J = 1$, choose a uniformly translated and oriented straight chain with length $\ell = 1$ with probability $1 - q$ and $\ell = 2$ with probability q , and set its two endpoints to one. Finally, pass every detector through an independent binary symmetric channel (BSC) with crossover ϵ .

For every detector i ,

$$\mathbb{P}_q(Y_i = 1) = \epsilon + (1 - 2\epsilon)\frac{2\eta}{m}, \quad (4)$$

which is independent of q . More strongly, conditional on $J = 1$ every clean template has Hamming weight two. Since iid BSC noise is invariant under coordinate permutations, the complete distribution of $C = \sum_i Y_i$ is independent of chain length and hence of q . Independent rounds therefore give an exact count-*process* no-go by Corollary 1, not merely a zero mean-gap observation.

Let $z_i = 1 - 2y_i$, $\bar{z} = m^{-1} \sum_i z_i$, and

$$g_\ell(y) = \frac{1}{2m} \sum_{v, o \in \{x, y\}} z_v z_{v+\ell e_o}. \quad (5)$$

Write $B_0(y)$ for the BSC probability given the all-zero clean template and $r = (1 - \epsilon)/\epsilon$. Define $a = (r + r^{-1})/2$ and $b = (r - r^{-1})/2$. Averaging a two-endpoint likelihood ratio over translations and orientations gives

$$K_\ell(y) = a^2 - 2ab\bar{z} + b^2 g_\ell(y). \quad (6)$$

The exact one-round likelihood is therefore

$$P_q(y) = B_0(y) [(1 - \eta) + \eta\{(1 - q)K_1(y) + qK_2(y)\}]. \quad (7)$$

Theorem 2 (translation sufficiency and signed gap).

In the declared spatial family, $(\bar{z}, g_1, g_2)$ is sufficient for q . Moreover, for two parameters q_0, q_1 ,

$$\begin{aligned} &(\mathbb{E}_{q_1} g_1 - \mathbb{E}_{q_0} g_1, \mathbb{E}_{q_1} g_2 - \mathbb{E}_{q_0} g_2) \\ &= \frac{2\eta(1 - 2\epsilon)^2}{m}(q_1 - q_0)(-1, +1). \end{aligned} \quad (8)$$

Proof: Equation (7) follows by writing the likelihood ratio for endpoints u, v as $(a - bz_u)(a - bz_v)$ and averaging. Its dependence on q is entirely through $(\bar{z}, g_1, g_2)$, proving sufficiency by factorization. For (8), independent readout flips multiply every clean two-point spin product by $(1 - 2\epsilon)^2$. Moving event mass $\eta(q_1 - q_0)$ from length one to length two decreases the matching length-one average and increases the matching length-two average; uniform translation/orientation supplies the factor $2/m$. ■

This sufficiency is model-specific. It does not say that three statistics are sufficient for a general surface-code circuit, arbitrary DEM, or hardware noise.

B. Temporal persistence drift

Let $H_t \in \{0, 1\}$ denote the chain-length state, with stationary $\mathbb{P}(H_t = 1) = q_0$, and define

$$X_t = \frac{H_t - q_0}{\sqrt{q_0(1 - q_0)}}. \quad (9)$$

The null uses iid H_t . Under the alternative,

$$\mathbb{P}(H_{t+1} = 1 \mid H_t) = q_0 + \kappa(H_t - q_0), \quad (10)$$

so the stationary one-cycle law is unchanged and $\mathbb{E}(X_t X_{t+1}) = \kappa$. Let $e_h(y)$ be the emission density at state h , $p_0(y) = \sum_h \pi_h e_h(y)$, and

$$a(y) = \mathbb{E}_0[X_t \mid Y_t = y]. \quad (11)$$

Theorem 3 (exact adjacent-pair likelihood ratio). For a stationary adjacent pair under (10), relative to two iid null cycles,

$$L_\kappa(y, y') = 1 + \kappa a(y)a(y'), \quad \mathbb{E}_0 L_\kappa = 1. \quad (12)$$

Proof: For a binary stationary state, the joint state mass can be written $\pi_h \pi_{h'} [1 + \kappa x_h x_{h'}]$, where x_h is the standardized state value. Multiplying by $e_h(y)e_{h'}(y')$, summing over h, h' , and dividing by $p_0(y)p_0(y')$ gives (12). Null expectation one follows by integrating the likelihood ratio, equivalently from $\mathbb{E}_0 a(Y) = 0$. ■

Nonoverlapping pairs give independent null e-factors and a deliberately pair-restricted alternative. They discard cross-block alternative correlations. A forward-filter HMM is retained as a model-aware ceiling, not misnamed as the same pair likelihood.

V. PREDICTABLE OBSERVABLE E-DETECTOR

A. Raw and variance-aware directions

Encode one syndrome round as the trace-one correlation state

$$R_t = \frac{z_t z_t^\top}{m}. \quad (13)$$

A raw AOC witness is the positive support of a projected difference between a past-only live estimate and the null state. Translation averaging reduces it to displacement or Fourier features. This raw direction targets a bounded one-shot expectation gap, analogous to restricted total-variation discrimination. It is not a quickest-change solution.

For an accessible feature vector $G_t = g(R_t)$ with exact null mean μ_0 and covariance V_0 , a predictable live estimate gives

$$\hat{\delta}_{t-1} = \hat{\mathbb{E}}_{t-1} G - \mu_0, \quad w_{t-1} = (V_0 + \lambda I)^{-1} \hat{\delta}_{t-1}. \quad (14)$$

This is regularized Hotelling/Fisher whitening. Locally, it is the least-squares projection of a likelihood score onto the accessible feature span. We call the resulting restricted implementation vAOC only to identify its provenance in this repository, not to claim new whitening mathematics.

Let $B(w)$ be a predictable deterministic bound satisfying $|w^\top(G_t - \mu_0)| \leq B(w)$ for every possible G_t , and set

$$s_t = \frac{w_{t-1}^\top(G_t - \mu_0)}{B(w_{t-1})} \in [-1, 1]. \quad (15)$$

When $B = 0$, set $s_t = 0$. For predictable $\beta \in [0, 1]$,

$$L_t(\beta) = 1 + \beta s_t \geq 0, \quad \mathbb{E}_0[L_t(\beta) \mid \mathcal{F}_{t-1}] = 1. \quad (16)$$

B. Shiryayev–Roberts recursion and guarantee

For a fixed grid $\mathcal{B}$, initialize $S_0^\beta = 0$ and update

$$S_t^\beta = (S_{t-1}^\beta + 1)L_t(\beta), \quad \bar{S}_t = |\mathcal{B}|^{-1} \sum_{\beta \in \mathcal{B}} S_t^\beta. \quad (17)$$

Stop at $\tau_\gamma = \inf\{t : \bar{S}_t \geq \gamma\}$.

Proposition 1 (exact-null average run length). Under (16) and standard integrability conditions,

$$\mathbb{E}_0 \tau_\gamma \geq \gamma. \quad (18)$$

Proof: For each β , $S_t^\beta - t$ is a null martingale because $\mathbb{E}_0(S_t^\beta \mid \mathcal{F}_{t-1}) = S_{t-1}^\beta + 1$. The mixture has the same property. Optional stopping at $\tau_\gamma \wedge n$ gives $\mathbb{E}_0(\tau_\gamma \wedge n) = \mathbb{E}_0 \bar{S}_{\tau_\gamma \wedge n} \geq \gamma \mathbb{P}_0(\tau_\gamma \leq n)$. If the stopping time is defective, its extended expectation is infinite; otherwise monotone convergence yields (18). This is the Shiryayev–Roberts/e-detector argument [13]. ■

The statistic $\bar{S}_t$ is an *e-detector*, not a unit-mean e-process: under the null its expectation grows as t . Equation (18) is an average-run-length guarantee, not a Ville bound on the probability of ever crossing γ . If the witness or tuning uses current/future test data, or if centering/bounds are invalid, (16) need not hold. Here μ_0 is analytic. Conditional on independent auxiliary data, covariance, ridge, and logistic-direction fitting only fix a bounded, predictable effect, so exact-null centering preserves (16). Estimating the null mean itself from finite data without uncertainty protection does not automatically preserve the guarantee.

Algorithm 1: Predictable accessible-observable e-detector

Input: exact or uncertainty-valid μ_0 ; independently fitted V_0 ; feature bank g ; predictable live-state rule; ridge λ ; betting grid $\mathcal{B}$; threshold γ .

Initialize: past-only estimator and $S_0^\beta = 0$ for every $\beta \in \mathcal{B}$.

For $t = 1, 2, \dots$:

1) Before seeing Y_t , form the raw support direction or (14) from $\mathcal{F}_{t-1}$.

2) Observe Y_t , compute $G_t = g(R_t)$, and normalize by a declared support bound to obtain (15).

3) Update every component by (17); alarm if $\bar{S}_t \geq \gamma$.

4) If no alarm occurs, update the live estimate using Y_t .

Output: stopping time, predictable witness history, and component evidence.

Fig. 1. The witness must be frozen before the current observation. Independent calibration and held-out tuning are part of the data budget.

C. A necessary restart-at-change correction

If a control is exactly blind, then $s_t = 0$, $L_t = 1$, and $S_t = t$. With threshold γ and changepoint $\nu < \gamma$, the naive quantity $\gamma - \nu$ would falsely credit the age of the SR clock as post-change evidence. The protocol was therefore amended after a reduced pilot and before the locked run:

- 1) the primary response restarts every method at the changepoint and measures restricted mean delay on the same post-change segment;
- 2) each restart delay is compared with same-horizon no-change restricted run length;
- 3) surveillance results separately report pre-change alarms and condition post-change delay on $\tau > \nu$; and
- 4) independent no-change streams remain the ARL/censoring audit.

This amendment prevents deterministic clock accumulation from masquerading as detection.

VI. LOCKED DESIGN AND NAMED-COMPARATOR CRITERION

The primary controlled model uses $L = 5$, $\eta = 0.65$, $\epsilon = 0.03$, and $q_0 = 0.35$. Spatial alternatives are $q_1 \in \{0.45, 0.55, 0.65\}$; temporal alternatives are $\kappa \in \{0.50, 0.75, 0.90\}$. Locked-test comparisons use identical paired streams, changepoints, physical-cycle horizons, and seeds. Calibration uses 4096 physical cycles per family. A spatial update costs one cycle and uses threshold 1000; a temporal pair update costs two cycles and uses threshold 500. Both target an average run length of at least 1000 physical cycles.

The named non-oracle comparator is a linear logistic direction on the same Fourier simplex/pair-simplex, affinely mapped into $[0, 1]$, centered at the analytic null, and passed through the same bounded-score SR. It was chosen after the independent offline audit but before the sequential locked test; it is not claimed to be the strongest generic baseline. Its training uses eight independent streams per class and 512

cycles per stream. vAOC receives the same $8 \times 512 = 4096$ alternative ridge-validation cycles plus 4096 null covariance-calibration cycles; the logistic comparator receives 4096 cycles from each class with the same stream structure.

The comparison set includes a provably blind detector-count control, mean-vector scans, matched signed displacement correlation, shrinkage Hotelling/covariance and spectral tests, RBF-Hamming MMD, raw AOC, vAOC, the named logistic comparator, direct observable-bank e-detectors, exact grid likelihood-ratio SR, and a known-post/HMM ceiling. Separate logistic-regression and RBF-SVM window classifiers are offline diagnostics without sequential guarantees.

At $q_1 = 0.55$ and $\kappa = 0.75$, let $D_i = T_i^{\text{vAOC}} - T_i^{\text{logit}}$. Restricted delays lie in $[0, H]$, hence independent paired differences lie in $[-H, H]$. Under $H_0 : \mathbb{E}D_i \geq 0$, the distribution-free one-sided Hoeffding p-value and Bonferroni simultaneous upper bound are

$$p_H = \min \left\{ 1, \exp \left[-\frac{n \max(0, -\bar{D})^2}{2H^2} \right] \right\}, \quad (19)$$

$$U_H = \bar{D} + H \sqrt{\frac{2 \log(m/\alpha)}{n}}, \quad m = 2. \quad (20)$$

Holm adjustment is applied to the two p-values. A paired percentile-bootstrap interval is descriptive only. For each task, $U_H < 0$, the Holm-adjusted p-value must be below 0.05, and the ARL condition must hold; the repo-level flag additionally requires both tasks to pass. Thus the tested contrast is

$$\text{RMDD}_{\text{vAOC}} - \text{RMDD}_{\text{named logistic comparator}}. \quad (21)$$

Even a pass supports only vAOC versus this named comparator on these two controlled tasks, not strongest-baseline or general algorithmic superiority. Exact likelihoods and full-HMM/known-post procedures are ceilings, not targets to beat. Failure is an informative negative result and does not alter the no-go theorems.

VII. FROZEN RESULTS

A. Exact identifiability certificates

The locked controlled audit used $L = 5$, $q_0 = 0.35$, $q_1 = 0.55$, $\eta = 0.65$, and $\epsilon = 0.03$. Table II separates exact identity residuals from finite-sample discrimination summaries. The spatial AUC used 200,000 samples per regime; the temporal AUC used 100,000 pairs per regime. The AUCs are not claims of strong practical performance; they verify that the richer observable contains nonzero information where the declared lower level is exactly blind.

B. Offline fixed-window diagnostic

The objective-corrected locked offline audit uses 800 training, 400 validation, and 1000 test windows per class. Spatial windows contain 64 cycles; temporal windows contain 64 nonoverlapping pairs, or 128 cycles. Table III reports the non-ceiling method selected by validation ROC AUC and its untouched test AUC. Detector-count sequence AUCs range only from 0.488435 to 0.524807, consistent with finite-sample fluctuation under the exact count-process no-go. Known-model likelihoods remain ceilings.

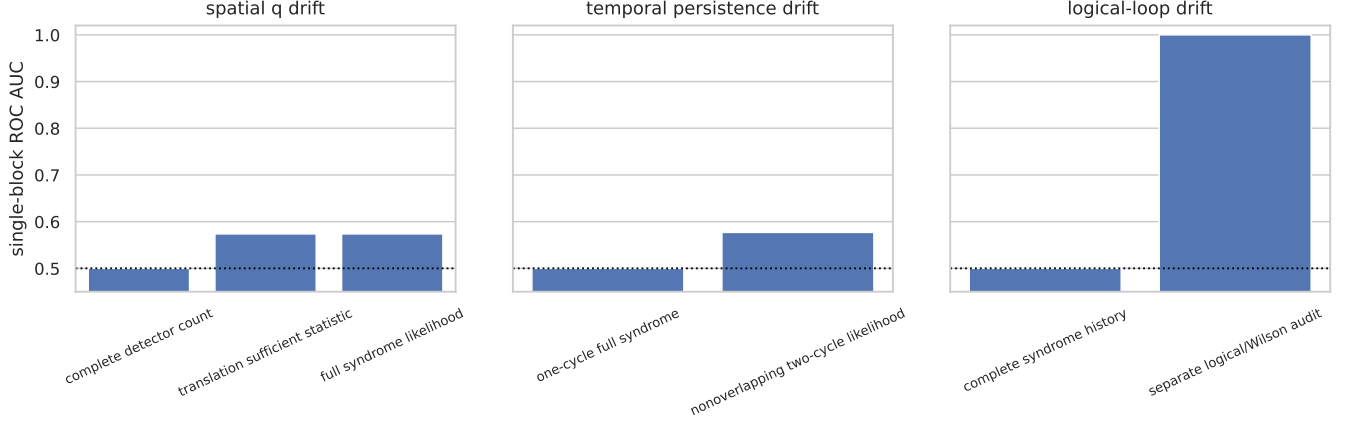


Fig. 2. Locked controlled-model access hierarchy. Count blindness is exact for the spatial iid family; one-cycle equality in the temporal family does not imply path blindness. The logical loop is syndrome-invisible and requires a separately costed logical audit.

TABLE II
CONTROLLED IDENTIFIABILITY AUDIT.

Quantity	Locked value
Complete count-PMF total variation	0
Maximum detector-marginal gap	0
Spatial sufficient/direct log-LR error	5.55×10^{-15}
Temporal pair closed-form/direct error	8.91×10^{-15}
Spatial exact-likelihood ROC AUC	0.573775
Temporal pair-likelihood ROC AUC	0.576994
Logical syndrome-history total variation	0
Syndrome-only / logical-audit success	0.5/1.0

TABLE III
LOCKED OFFLINE TEST ROC AUC. “D4” DENOTES THE DIHEDRAL
FOURIER/PAIR-LIFT BANK; SELECTION USED VALIDATION LABELS ONLY.

Family	Effect	Selected non-ceiling	Test AUC	Ceiling
Spatial	$q = .45$	Translation logistic	.77936	.79618
Spatial	$q = .55$	D4 Hotelling	.92187	.93572
Spatial	$q = .65$	D4 Hotelling	.98452	.98984
Temporal	$\kappa = .50$	D4 logistic	.84943	.96487
Temporal	$\kappa = .75$	D4 logistic	.94668	.99925
Temporal	$\kappa = .90$	D4 RBF SVM	.97499	.99999

An independent objective audit corrected the estimator interpretation before this frozen rerun. Only the indicator of the positive support on the D4 Fourier/pair-lift probability simplex is called an AOC effect. Positive-part directions on count, first-moment, or translation features are generic linear mean contrasts, not Helstrom or AOC optima. At the two middle effects, the simplex positive-support projector obtains AUC 0.67465 spatially and 0.51848 temporally, compared with 0.92187 and 0.76853 for same-bank regularized Hotelling and 0.92216 and 0.94668 for same-bank logistic. This offline benchmark has no sequential or ARL guarantee.

C. Measurement-policy audit

For 100 declared ZZ displacement observables on the $L = 5$ diagonal controlled states, the local-Pauli shadow estimator is

$$\widehat{\langle Z_i Z_j \rangle} = 9 \mathbf{1}\{B_i = B_j = Z\} b_i b_j. \quad (22)$$

Only 1/9 of copies contribute to a given pair, whereas native all- Z readout contributes every copy. Across 256 repetitions at 4096 copies, the mean RMSEs were 0.045054 (shadow) versus 0.010737 (native) at $q = 0.35$, and 0.045192 versus 0.010674 at $q = 0.65$. The ratios are 4.20 and 4.23.

D. Circuit-level decoder utility

A separate locked circuit experiment generated `surface_code:rotated_memory_z` circuits at distances $d \in \{3, 5, 7\}$ with rounds equal to distance, using Stim 1.16.0 and PyMatching 2.4.0. Each reference/correlated regime used 200,000 shots per distance. The circuit noise parameters were post-Clifford depolarization 0.001, measurement/reset flips 0.002, marginal data- X probability 0.02, and common paired data- X probability 0.01. An exact DEM audit found a maximum reference–correlated detector-marginal gap 3.33×10^{-16} , below the declared 10^{-12} tolerance.

At the predeclared distance-seven correlated endpoint, the static, ordinary-post, and correlation-aware logical-error rates were 0.01653, 0.01526, and 0.01120. The paired difference between ordinary-post and correlation-aware decoding was 0.00406, with paired-bootstrap 95% interval $[0.00369, 0.004435]$, a 26.6% relative reduction. Under the reference regime, however, the same three rates were 0.019785, 0.021605, and 0.022175: applying the correlation-aware model when it is wrong degrades performance.

This result supports downstream utility of correct correlation modeling under the specified simulator. It does not show that AOC/vAOC detects the change, does not compare a full DGR reproduction, and is neither hardware evidence nor an oracle-beating result.

E. Cycle-fair sequential detection

The publication-grade sequential run uses 512 null streams through 5000 cycles and 256 paired changed streams per effect, with change time 256 and a 1024-cycle post-change restriction horizon. Ridge selection on independent streams

TABLE IV

LOCKED MIDDLE-EFFECT SEQUENTIAL RESULTS IN PHYSICAL CYCLES. NRL IS THE NULL RESTRICTED MEAN RUN LENGTH THROUGH 5000 CYCLES; RMDD IS RESTART-AT-CHANGE RESTRICTED MEAN DELAY THROUGH 1024 CYCLES, WITH MISS FRACTION IN PARENTHESES. “GRID” USES THE EXACT SPATIAL LIKELIHOOD AND THE FULL-HMM TEMPORAL LIKELIHOOD.

Method	Spatial $q_1 = .55$			Temporal $\kappa = .75$		
	NRL	RMDD	Miss	NRL	RMDD	Miss
DFR/count	1000.0	1000.0	.000	1000.0	1000.0	.000
Model-aware grid	1169.8	88.3	.000	1364.4	53.2	.000
Matched witness	1001.9	459.8	.000	1001.1	963.7	.000
Raw AOC	1035.2	592.8	.031	992.5	862.9	.453
vAOC/Hotelling	1009.4	540.3	.000	1000.9	959.4	.297
Named logistic	1001.6	470.1	.000	1002.9	968.5	.398

freezes $\lambda = 10^{-3}$ spatially and 10^{-4} temporally. For the spatial family one detector update is one physical cycle and the threshold is 1000; for the temporal family one update is a nonoverlapping two-cycle pair and the threshold is 500. Both therefore target ARL at least 1000 physical cycles.

The exactly blind DFR control has zero restart-delay reduction relative to its same-horizon no-change run length. At the spatial endpoint, the corresponding reductions are 637.21, 495.88, 276.75, 403.68, and 486.92 cycles for the likelihood grid, matched witness, raw AOC, vAOC, and logistic, respectively. At the temporal endpoint they are 694.64, 37.38, 3.07, 18.50, and 2.73 cycles. The exact pair-restricted temporal likelihood has RMDD 110.91; the known-post one-cycle spatial and full-HMM temporal ceilings have RMDD 86.82 and 51.84. These controls confirm that the injected changes are detectable when the correct model is supplied; they are not proposed-method victories.

The preregistered secondary delay summaries tell the same story. Spatially, the vAOC IQR is [473, 591.25] cycles and the logistic IQR is [432.75, 501]; neither detects within 64 or 128 cycles, whereas the model-aware grid fractions are 0.3477 and 0.8242. Temporally, the vAOC and logistic IQRs are [914, 1024] and [934, 1024], and both have zero within-64/128 fractions; the full-HMM grid fractions are 0.7109 and 0.9844. Miss fractions appear in Table IV.

For the named comparison, the spatial mean difference $\bar{D} = \text{RMDD}_{\text{vAOC}} - \text{RMDD}_{\text{logit}}$ is 70.176 cycles: vAOC is slower. Its descriptive paired-bootstrap interval is [61.937, 78.672], while the predeclared simultaneous one-sided Hoeffding upper bound is 244.013 and the Holm-adjusted p-value is 1. Temporally, $\bar{D} = -9.180$ cycles, but the descriptive interval [-21.758, 3.148] crosses zero; the simultaneous upper bound is 164.657, the raw one-sided Hoeffding p-value is 0.9898, and the Holm-adjusted value is 1. Both fixed effects retain the analytic-null ARL guarantee by design, and their empirical null restricted mean run lengths have zero censoring. Nevertheless, neither task passes the statistical criterion, so the two-hypothesis overall flag is false. The locked experiment therefore gives *no support* for vAOC over the named validation-trained logistic effect on either predeclared comparison or overall.

TABLE V

LOCKED SPATIAL SCALING. DELAYS ARE RESTART RMDD; TIMINGS ARE SINGLE-PROCESS SECONDS PER 10,000 SAMPLES AFTER WARM-UP.

L	Gap _{max}	vAOC	Matched	Known post	Feat. s	LR s
5	.009189	542.25	462.38	85.02	.00292	.03527
7	.004688	744.07	636.72	85.51	.01579	.16245
9	.002836	867.81	740.40	99.89	.00897	.12271

F. Spatial scaling audit

The scaling arm freezes the $L = 5$ ridge, uses 128 paired repetitions per size, and retains $q_1 = .55$. Table V shows the analytic translation gap shrinking as $1/L^2$ and all non-oracle correlation-witness delays worsening. The likelihood ceiling remains much faster because it is given the exact alternative.

Raw-AOC delays are 608.77, 719.54, and 827.98 cycles for $L = 5, 7, 9$; count remains at 1000. These wall-clock measurements are an engineering audit, not a complexity theorem. They provide no computational or sample-efficiency scaling advantage. Separate `tracemalloc` calls give peak traced allocations of 7.79/19.51, 15.11/37.82, and 24.88/62.23 MiB for translation features/exact likelihood at $L = 5, 7, 9$. These are Python/NumPy allocation peaks, not total process RSS or a memory-complexity theorem.

Run 5 locked offline diagnostic audit

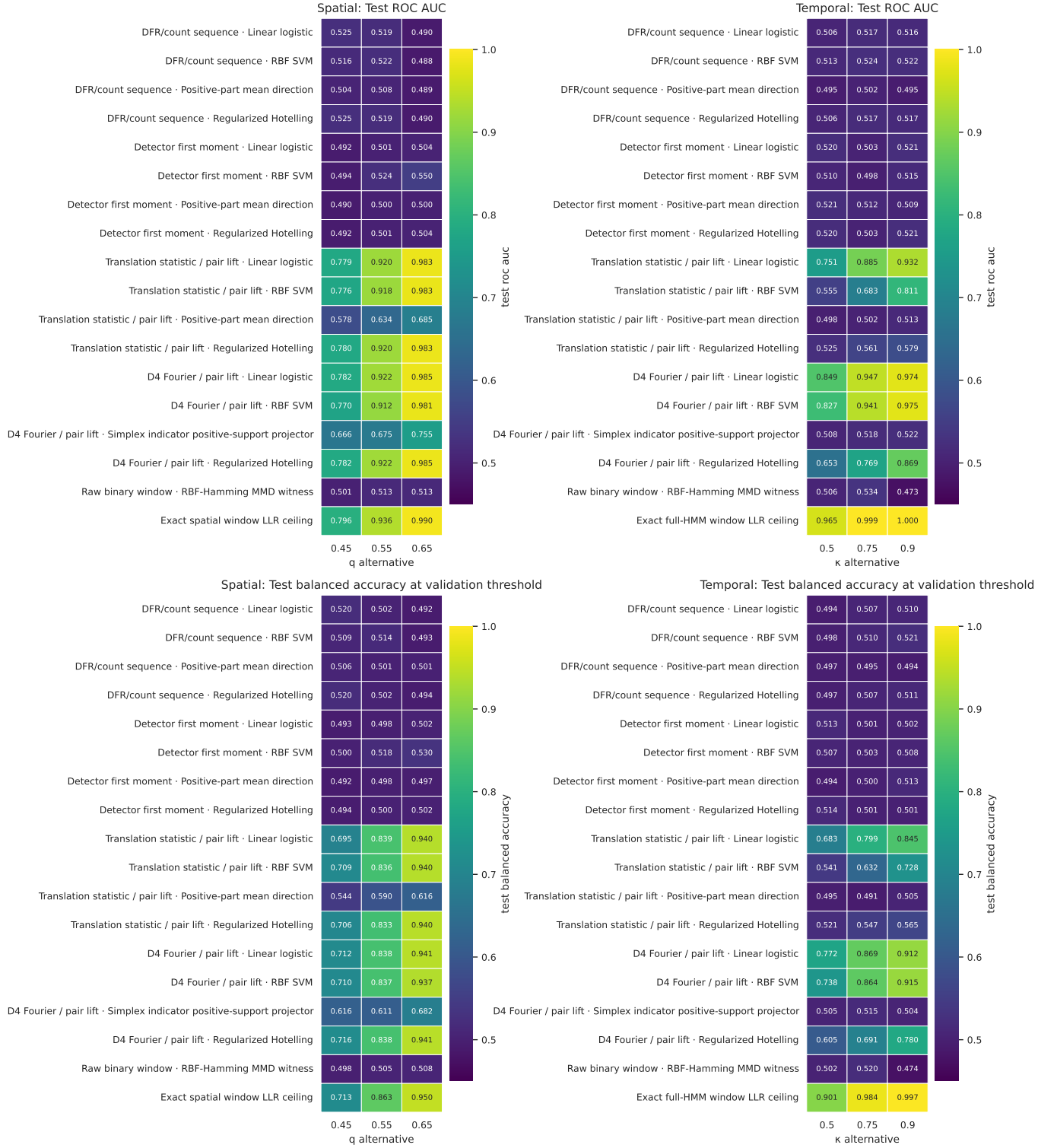


Fig. 3. Objective-corrected locked offline audit. Structured feature banks recover the controlled correlation changes, while count sequences remain at chance. Exact spatial and full-HMM likelihoods are model-aware ceilings.

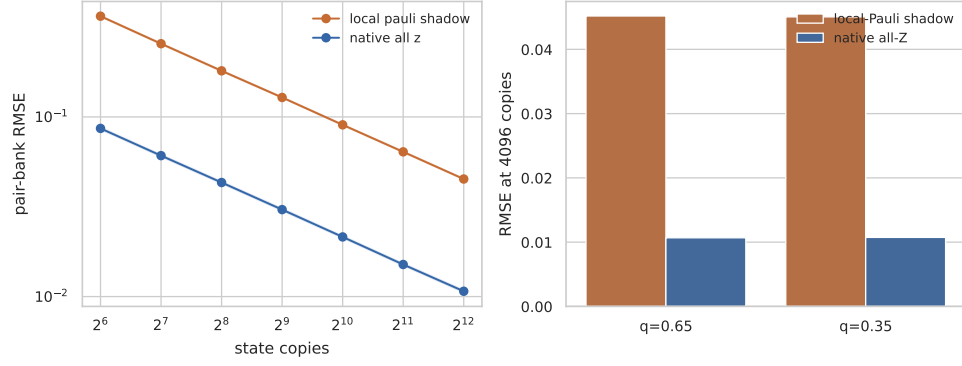


Fig. 4. Locked measurement-policy audit. Native all- Z is matched to the declared diagonal ZZ bank; the shadow penalty measures universality cost. This is not an implementation of eSCD and does not establish superiority to classical-shadow sequential detection.

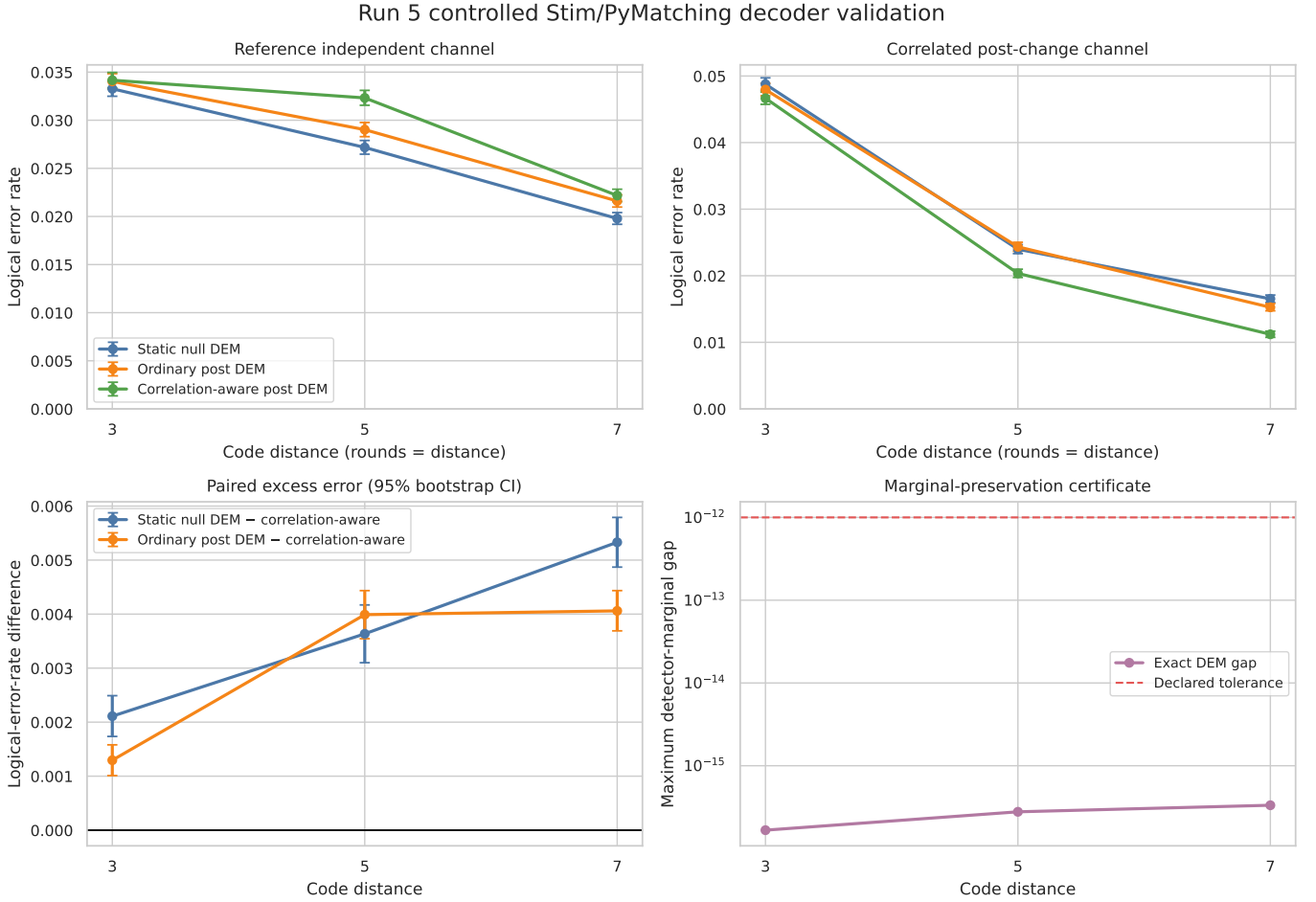


Fig. 5. Locked Stim/PyMatching circuit-level simulation. Correlation-aware decoding helps only in the injected correlated regime and is harmful under the reference regime, motivating reliable model selection. The injected channel and decoder models are known; no online detector is tested here.

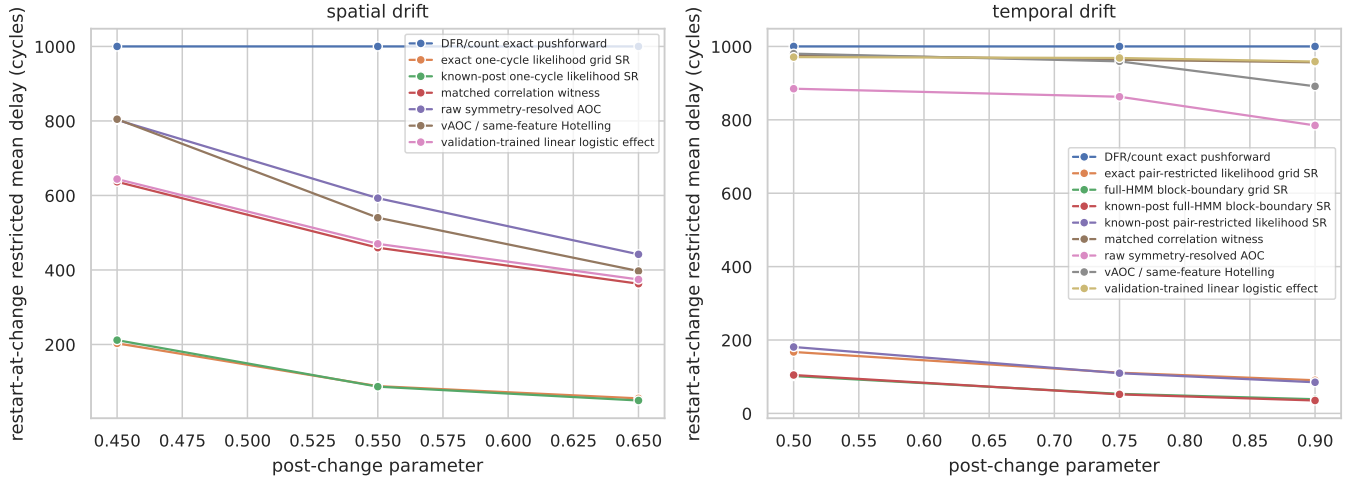


Fig. 6. Cycle-fair locked sequential audit. Delays use fresh starts at the change; no-change and surveillance experiments are kept separate. The temporal threshold is counted in pair updates but all displayed budgets and delays are physical cycles.

VIII. WHAT THE EVIDENCE DOES AND DOES NOT SHOW

The exact controlled results establish an information-access separation: count compression can be provably blind while declared spatial pair observables remain informative; one-cycle equality can coexist with detectable temporal dependence; and a logical cycle can be pathwise invisible to all syndrome-only rules. They do not establish universal sample efficiency, realistic device completeness, or asymptotic scalability.

The offline and sequential results separate feature access from algorithmic advantage. Structured features recover information unavailable to count compression, but the AOC projector is not the strongest offline rule and vAOC does not beat the named validation-trained logistic effect under the predeclared sequential inference. This is a negative result for that named comparison, not a theorem that vAOC is inferior for all distributions, budgets, or access restrictions.

The circuit result establishes that correct correlation-aware decoder modeling can lower logical error in one known injected channel. It also shows why false model selection matters. Because no online alarm-and-update pipeline is used in that arm, it is not evidence that the proposed detector produces the utility gain. Existing DGR, syndrome-to-DEM estimation, logical-error estimation, and Bayesian tracking remain mandatory comparators for a faithful deployment claim [1]–[4].

The following claims are explicitly prohibited by the current evidence:

- superiority over Helstrom discrimination, a known-post likelihood/HMM oracle, a supplied Wilson/logical audit, or a threshold/logistic rule given the identical known parity feature;
- quantum acceleration, a universal sample-efficiency advantage, or a scalable computational advantage;
- the first syndrome drift detector, syndrome noise estimator, correlation-aware decoder, covariance detector, or e-detector;
- a faithful performance comparison with eSCD from the shadow measurement-policy audit alone;
- hardware performance, broad fault-tolerance improvement, or robustness to arbitrary coherent, leakage, non-Markovian, and scheduling effects; and
- discovery of Wilson loops, local topological order, a toric/surface-code theorem, string theory, or holographic duality.

IX. LIMITATIONS AND NEXT EXPERIMENTS

First, exact analytic-null centering gives the displayed ARL theorem. Conditional on independent auxiliary data, covariance, ridge, and logistic fitting preserve it because they only fix a bounded predictable effect. Finite estimation of the null mean without uncertainty protection generally does not; confidence-valid centering or separate threshold calibration is then required. Second, the controlled model has deliberately low-dimensional sufficient statistics. Exact likelihood and full-HMM methods win here when the model is known; a constrained witness is valuable only if it remains competitive under model uncertainty, interpretability, or computational

restrictions. Third, the circuit arm uses stationary reference and correlated circuits rather than a within-run changepoint and knows the injected DEM. A complete practical test must couple detection, independent post-alarm estimation, decoder update, and held-out logical outcomes at a common total-shot budget.

Fourth, native syndrome readout is already a task-specific measurement. The shadow audit says only that measurement universality is costly for a known diagonal observable bank; universal shadows remain useful when future observables are not known in advance. Fifth, distances three through seven and one circuit family do not support a scaling theorem.

The preregistration also listed “witness overlap/localization” but did not freeze a target witness, overlap formula, or time aggregation for the dynamic vAOC history, and the locked run did not persist that history. We therefore do not manufacture a post-hoc localization number. This secondary item remains unfulfilled, supports no localization claim, and does not alter the predeclared paired-delay decision.

The string/gauge extension is therefore framed as a concrete future experiment, not a solved theory problem: in a lattice gauge model or tensor network, declare a gauge-invariant regional algebra, compare its reduced states before and after a controlled change, and resolve charge/flux sectors while recording the algebraic center. Such an experiment could test whether the same access hierarchy is diagnostically useful. It would not by itself establish a continuum string model or holographic duality.

X. CONCLUSION

The central negative result is simple but easy to overstate: sequential blindness follows from equality of the *entire accessible process law*. One-round equality suffices only with an additional product-law condition. The controlled surface-code-inspired family turns that distinction into three executable cases: exact count blindness with spatial witness recovery, one-cycle blindness with temporal-pair recovery, and a pathwise syndrome-invisible logical cycle.

A predictable bounded contrast can be used inside an established e-detector and inherits an exact-null average-run-length guarantee. Locked identifiability checks validate the analytic likelihood reductions; the measurement audit quantifies the cost of universal local-Pauli readout for a known ZZ bank; and circuit-level simulation demonstrates the downstream value, and the risk, of correlation-aware decoding under a known channel. The corrected offline audit favors standard logistic/Hotelling rules over the simplex positive-support projector, and the cycle-fair sequential test gives no support for vAOC over the named validation-trained logistic effect on either predeclared task or overall. Accordingly, Run 5 establishes a reproducible information-access benchmark and an honest negative named-comparator result, not a general algorithmic, quantum, sample-efficiency, topological, or string-theoretic advantage.

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